

TOMPUDA, Russian Federation

WMO: 304390

Lat: 55.117N

Lon: 109.750E

Elev: 465

StdP: 95.87

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.6	-30.1	-35.2	0.1	-32.3	-32.5	0.2	-29.7	11.1	-14.2	9.3	-11.5	0.6	10	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.5	23.5	15.8	21.5	15.3	19.6	15.2	17.7	20.3	16.9	19.6	16.0	18.7	1.6	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.9	12.8	18.5	15.9	11.9	17.9	14.9	11.2	17.1	52.0	20.8	49.3	19.7	46.7	18.8	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.5	6.5	4.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.7	-19.7	-19.2	-11.6	-2.0	4.2	10.1	14.7	14.8	8.2	0.1	-8.1	-12.9
	DBStd	12.75	5.82	6.02	5.92	3.98	3.13	3.29	3.36	2.45	3.80	3.71	5.50	5.42
	HDD10.0	4631	921	818	671	359	180	38	3	1	79	308	543	709
	HDD18.3	7317	1179	1051	929	609	437	248	121	111	303	567	793	968
	CDD10.0	367	0	0	0	0	2	40	148	151	26	0	0	0
	CDD18.3	11	0	0	0	0	0	0	8	2	0	0	0	0
	CDH23.3	65	0	0	0	0	0	6	49	11	0	0	0	0
	CDH26.7	8	0	0	0	0	0	0	7	1	0	0	0	0
Wind	WSAvg	1.6	1.4	1.0	1.4	1.6	1.5	1.2	1.2	1.3	1.7	1.9	2.3	2.2
Precipitation	PrecAvg	367	15	13	11	18	27	44	58	64	45	30	24	18
	PrecMax	473	57	25	34	42	69	92	136	107	82	71	49	32
	PrecMin	248	4	4	1	5	8	13	12	21	4	0	10	10
	PrecStd	61	11	5	8	9	13	23	33	22	17	17	10	7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.8	-3.7	3.3	11.0	17.0	24.3	27.4	24.9	20.6	10.2	5.5	2.4
		MCWB	-4.0	-5.0	0.9	5.2	9.1	15.4	17.9	16.6	14.0	6.5	3.6	1.2
	2%	DB	-6.7	-6.2	1.2	7.1	13.7	21.2	24.5	22.3	17.4	7.9	3.6	-1.5
		MCWB	-7.7	-7.0	-0.6	2.6	7.4	13.4	16.5	15.8	13.5	5.0	1.8	-3.0
	5%	DB	-9.5	-8.3	-1.0	4.7	11.5	18.5	22.3	20.5	15.8	6.4	1.6	-3.5
		MCWB	-10.2	-9.2	-2.8	1.6	5.9	12.6	16.0	15.6	12.5	3.7	-0.2	-4.9
	10%	DB	-11.8	-10.5	-2.9	3.3	9.4	15.6	20.1	19.1	14.0	5.2	-0.2	-5.3
		MCWB	-12.4	-11.2	-4.5	1.0	5.1	11.5	15.5	15.8	11.2	2.8	-1.9	-6.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-3.7	-4.8	1.2	5.6	10.3	16.0	19.8	18.6	15.7	7.4	3.8	0.7
		MCDB	-2.8	-3.7	2.3	10.2	15.4	21.8	22.5	19.8	18.1	9.9	5.3	1.8
	2%	WB	-7.6	-7.1	-0.6	3.4	8.1	14.3	18.0	17.7	14.4	5.6	1.9	-2.7
		MCDB	-6.8	-6.2	1.0	6.1	12.2	19.7	21.2	19.6	16.8	7.4	3.4	-1.6
	5%	WB	-10.4	-9.1	-2.6	2.3	6.9	13.1	17.1	16.9	13.1	4.1	0.0	-4.9
		MCDB	-9.8	-8.4	-1.1	4.2	10.4	17.2	20.6	19.0	15.2	6.0	1.6	-3.6
	10%	WB	-12.4	-11.3	-4.5	1.4	5.6	12.0	16.2	16.2	11.3	3.0	-1.8	-6.6
		MCDB	-11.9	-10.7	-3.0	3.1	8.6	15.4	19.3	18.4	13.8	4.7	-0.3	-5.3
Mean Daily Temperature Range	5% DB	MDBR	7.7	8.6	9.6	7.9	7.9	8.1	7.7	7.5	7.5	6.7	6.4	6.7
		MCDBR	7.6	8.8	9.9	9.8	11.9	13.7	12.3	10.2	8.7	8.4	6.7	7.1
		MCWBR	6.7	7.9	8.2	6.4	6.8	7.2	5.9	5.5	5.1	6.1	6.5	6.7
	5% WB	MCDBR	7.2	8.7	9.0	8.5	10.7	11.3	9.0	7.4	7.6	8.0	6.7	7.0
		MCWBR	6.6	7.9	7.4	6.2	6.9	6.9	5.7	5.0	5.1	6.5	6.8	6.8
Clear-Sky Solar Irradiance	taub		0.250	0.284	0.309	0.352	0.395	0.433	0.417	0.384	0.335	0.294	0.262	0.242
	taud		2.081	2.065	2.087	2.141	2.172	2.223	2.323	2.392	2.462	2.352	2.107	2.035
	Ebn at Noon		669	778	851	865	846	815	822	830	831	777	641	577
	Edh at Noon		64	95	119	132	136	132	117	103	83	72	61	53
All-Sky Solar Radiation	RadAvg		0.90	1.87	3.38	4.66	5.30	5.81	5.29	4.41	3.02	1.71	0.90	0.63
	RadStd		0.07	0.12	0.16	0.32	0.39	0.38	0.52	0.30	0.24	0.17	0.09	0.05

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page